

YASWANTH KOSURU

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SUMMARY

Full-stack engineer specializing in AI-integrated web applications across Javascript and Python stacks. Open-source contributor with 6+ PRs merged into stdlib-js and Zulip. Proven in shipping 0-to-1 products, designing event-driven microservices, and building production LLM/RAG pipelines with Langchain and Langgraph.

WORK EXPERIENCE

Full Stack Developer — Learnfluid

Apr 2025 – Present

Remote — React.js, Node.js, Prisma, Langchain, Langgraph, PostgreSQL, Vercel

- Replaced a manual Excel-based tutoring workflow by shipping two production platforms (tutor + student) end-to-end, covering class scheduling, content delivery, and a multi-stage approval workflow for subject-lead content review.
- Engineered a Langchain + Langgraph RAG pipeline that auto-generates quizzes, flashcards, and notes from uploaded course material, eliminating manual content-creation effort per topic.
- Built a content authoring engine supporting 10+ interactive question formats (MCQ, true/false, match, fill-in-blank, etc.) with tutor-attachable workflows, enabling reusable lesson assets across all scheduled classes.

Backend Developer — Framespik — Freelance

Dec 2024 – Feb 2025

Remote — Django, Django REST Framework, Django Channels, Redis, AWS S3, Azure Blob

- Delivered 15+ REST API endpoints for user onboarding, multi-cloud asset uploads (AWS S3 + Azure Blob), and crew management using Django and DRF from the ground up.
- Implemented real-time UI customization with Django Channels and Redis Pub/Sub, synchronizing state across concurrent user sessions over WebSocket connections.

Full Stack Developer Intern — A.I HYR

Mar 2024 – Jul 2024

Hyderabad — Node.js, Express, React, MongoDB, Azure Functions, Azure Queue, Azure Blob, Python, OpenAI API

- Scaled concurrent upload capacity 50x (20 to 1,000+ simultaneous sessions) by re-architecting video/audio ingestion from backend-proxied transfers to direct client-to-Azure-Blob multipart uploads with server-side block commits, cutting per-session server memory consumption from 1–2 GB to near zero.
- Eliminated cascading API failures during bulk resume processing (1,000+ concurrent evaluations) by decoupling synchronous OpenAI calls into two Azure Queue pipelines — a blob-triggered serverless function for NLP-based name extraction and a timer-triggered queue for rate-controlled LLM evaluations.
- Expanded the platform's addressable market from recruiters to students by designing and shipping candidate-facing interview-prep flows and partner management dashboards, enabling onboarding of 2+ educational partners and 500+ student users within the first 1 months post-launch.

PROJECTS

Self-Hosted Code Execution Platform (LeetCode-style Judge)

 [GitHub](#)

- Engineered a sandboxed code-execution platform running untrusted programs inside Firecracker microVMs with vsock-only communication, achieving **29ms p50 latency**, **15.8 req/s** at 4 concurrent users, and zero failures under load.
- Built a judging pipeline distributing 100+ test cases across a warm microVM pool with real-time NDJSON verdict streaming, supporting 6 languages (Python, C++, C, Java, Go, Bash) with per-language resource limits and timeout enforcement via cgroups.
- Developed a contest workspace in Next.js and React with Monaco Editor integration and live verdict streaming over Server-Sent Events.

Distributed WebSocket Chat System

 [GitHub](#)

- Architected a horizontally scalable WebSocket chat system using Node.js servers + Redis Pub/Sub channels keyed by chat room, decoupling pods through subscription-based fan-out so new nodes scale linearly without inter-pod coordination or message loss.
- Deployed the platform on Kubernetes with horizontal pod autoscaling, liveness/readiness probes for automatic failure recovery, and rolling updates for zero-downtime releases — all containerized via Docker.
- Instrumented observability with Prometheus metrics (WebSocket connection count, message throughput, broker lag) feeding Grafana dashboards for real-time visibility into latency, throughput, and error rates.

EDUCATION

2019 - 2023 B.Tech in Electronics & Communication Engineering at **Sagi Rama Krishnam Raju Engineering College** (CGPA: 8.00/10.00)

OPEN SOURCE CONTRIBUTIONS

stdlib-js — 6+ PRs Merged

 github.com/stdlib-js

- Shipped 4 production utility modules to stdlib-js (JavaScript standard library for numerical and scientific computing) — **cuany-by** (conditional cumulative operations), **cusome** (threshold-based transforms), **is-array-like-object** (type guard), and **fs/read-ndjson** (NDJSON file reader) — all reviewed and merged by core maintainers.
- Optimized runtime performance and expanded test coverage across existing array and utility modules, with refactoring contributions adopted into the production library.

SKILLS

Languages	JavaScript, TypeScript, Python, SQL
Frontend	React.js, Next.js, Redux, Tailwind CSS, HTML/CSS
Backend	Node.js, Express.js, Django, Django REST Framework, Django Channels, FastAPI, Prisma
Databases	PostgreSQL, MongoDB, Redis, MySQL
Cloud / DevOps	Azure (Functions, Queue, Blob Storage), AWS S3, Docker, Kubernetes, Vercel, Prometheus, Grafana
AI / LLM	Langchain, Langgraph, OpenAI API, RAG Pipelines, Python NLP
Systems	Distributed Systems, Microservices, Design patterns, Web Sockets, REST API Design